作业

作业题目来自

梁昆淼《力学》 第 4 版 高等教育出版社

例： L01.08 梁昆淼书第 01 章第 08 题

Johns Analytical Mechanics 2nd ed Oxford University Press

例： J02.05 Johns 书第 02 章第 05 题

J03.07† 带 † 表示仅要求荣誉班同学完成

Homework 07, Week 09 2023.10.30 Due 2023.11.08 L08.04, L08.05, J05.06, J05 .09†, J15.02

★ L08.04

试应用光学中的费马原理研究光线在 x-o-y 平面上的传播，已知折射率 n 反比于坐标 y

Solution

一般参数法：

$$T=\int_{\beta_1}^{\beta_2}\frac{ds}{c/n}=\frac{\alpha}{c}\int_{\beta_1}^{\beta_2}\frac{\sqrt{\dot{x}^2+\dot{y}^2}}{y}d\beta=\frac{\alpha}{c}\int_{\beta_1}^{\beta_2}f(x,y,\dot{x},\dot{y})\,d\beta,\qquad \text{其中}\ f(x,y,\dot{x},\dot{y})=\frac{\sqrt{\dot{x}^2+\dot{y}^2}}{y}$$

取驻值路线的充要条件是

$$\frac{d}{d\beta}\frac{\partial f}{\partial\dot{x}}-\frac{\partial f}{\partial x}=0\ \Longrightarrow\ \frac{\dot{x}}{y\sqrt{\dot{x}^2+\dot{y}^2}}=c_1,\qquad x\text{ 是可遗坐标，对应于一个运动常数}\qquad(1)$$

$$\frac{d}{d\beta}\frac{\partial f}{\partial\dot{y}}-\frac{\partial f}{\partial y}=0\ \Longrightarrow\ \frac{d}{d\beta}\frac{\dot{y}}{y\sqrt{\dot{x}^2+\dot{y}^2}}+\frac{\sqrt{\dot{x}^2+\dot{y}^2}}{y^2}=0\qquad(2)$$

取 $\beta=x$，并记得 f 是 $\dot{x}$, $\dot{y}$ 的齐一次函数，以上两个方程仅一个独立，取第一个方程得

$$\frac{1}{y\sqrt{1+y'^2}}=c_1\quad\Longrightarrow\ y'=\frac{\sqrt{1-c_1^2y^2}}{c_1y}\quad\Longrightarrow\ dx=\frac{c_1y\,dy}{\sqrt{1-c_1^2y^2}}$$

$$\Longrightarrow\quad x-c_2=\sqrt{(1/c_1^2)-y^2}\quad\Longrightarrow\ (x-c_2)^2+y^2=(1/c_1)^2\quad\text{当 }c_1=0\text{ 时退化为}\quad x=c_2$$

坐标参数法：

$$T=\int_{x_1}^{x_2}\frac{ds}{c/n}=\frac{\alpha}{c}\int_{x_1}^{x_2}\frac{\sqrt{1+y'^2}}{y}dx\qquad\text{其中}\ f(x,y,y')=\frac{\sqrt{1+y'^2}}{y}$$

取驻值路线的充要条件是

$$\frac{d}{dx}\frac{\partial f}{\partial y'}-\frac{\partial f}{\partial y}=0\ \Longrightarrow\ \frac{d}{dx}\frac{y'}{y\sqrt{1+y'^2}}+\frac{\sqrt{1+y'^2}}{y^2}=0\qquad(3)$$

接着，利用第二形式的欧拉-拉格朗日方程（E-L 方程），类似于广义能量定理：

$$\text{定义：}\ h=\sum_{k=2}^{N}x_k'\frac{\partial f(x,x_{[1]}')}{\partial x_k'}-f(x,x_{[1]}'),\quad E\text{-}L\text{ 方程：}\ \frac{dh}{dx_1}=-\frac{\partial f(x,x_{[1]}')}{\partial x_1}$$

$$\text{可得：}\ h=y'\frac{df}{dy'}-f=-\frac{1}{y\sqrt{1+y'^2}},\qquad\text{由E-}L\text{ 方程：}\ \frac{dh}{dx}=-\frac{\partial f}{\partial x}=0$$

$$\Longrightarrow \quad -\frac{1}{y\sqrt{1+y'^2}} = -c_1 \tag{4}$$

式 (3) 与 $\beta = x$ 时的 式 (2) 相同，式 (4) 与 $\beta = x$ 时的 式 (1) 相同。

从坐标参数法可知，(3) 和 (4) 并不独立（正如广义能量定理与拉格朗日方程不独立）

对应于一般参数法，(1) 和 (2) 也就不应该是独立的，

如何在坐标参数法中给予证明 —— 利用 $f = (1/y)(\dot{x}^2 + \dot{y}^2)^{1/2}$ 是 $\dot{x}$ 和 $\dot{y}$ 的齐一次函数。

★ L08.05

在 x-o-y 平面上给定两点，试用曲线把两点相连，使连线绕 x 轴旋转所形成的曲面面积最小

Solution

坐标参数法：

$$S = \int_A^B 2\pi y\, ds = 2\pi \int_A^B y\sqrt{1+y'^2}\, dx \qquad f = y\sqrt{1+y'^2}$$

注意这里 f 与 x 无关，故可利用第二形式的欧拉-拉格朗日方程（类似于广义能量定理）。

$$h = \sum_{k=2}^{N} x'_k \frac{\partial f(x, x'_{[1]})}{\partial x'_k} - f(x, x'_{[1]}) = y'\frac{\partial f}{\partial y'} - f = -\frac{y}{\sqrt{1+y'^2}}$$

$$\frac{dh}{dx} = -\frac{\partial f}{\partial x} \quad \Longrightarrow \quad \frac{y}{\sqrt{1+y'^2}} = c_1 \quad \Longrightarrow \quad \frac{dy}{\sqrt{(y/c_1)^2 - 1}} = dx \quad \Longrightarrow \quad c_1 \operatorname{arccosh}\left(\frac{y}{c_1}\right) = x - c_2$$

$$\Longrightarrow \quad y = c_1 \cosh\left(\frac{x - c_2}{c_1}\right), \quad \text{其中 } c_1 \text{ 和 } c_2 \text{ 由两端点的坐标确定}$$

上式类似于广义能量定理 $\frac{dH}{dt} = -\frac{\partial L}{\partial t}$，这里看不大惯是因为 t 变成 x 了，把 x 看成参数就会习惯些。

一般参数法：

$$S = \int_A^B 2\pi y\, ds = 2\pi \int_A^B y\sqrt{\dot{x}^2 + \dot{y}^2}\, d\beta \qquad f = y\sqrt{\dot{x}^2 + \dot{y}^2}$$

f 是 $\dot{x}$ 和 $\dot{y}$ 的齐一次函数，两个方程不独立，选其中简单的方程求解。方程形式：$\frac{d}{d\beta}\frac{\partial f}{\partial \dot{x}_k} - \frac{\partial f}{\partial x_k} = 0$

显然 $k = 1$，$x_k = x$ 方程较为简单：$\frac{d}{d\beta}\left(\frac{y\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}}\right) = 0 \xRightarrow{\text{取 } \beta = x} \frac{y}{\sqrt{1+y'^2}} = c_1$

接着求解这个微分方程，求解过程同上。

★ J05.06

A right-circular cylinder of top radius a is oriented with its symmetry axis along $\hat{e}_3$.

(a) Use the culculus of variations and cylindrical polar coordinates to find the two differential equations describing the extremum path between two general points on the surface of the cylinder.

(b) Choose β equal to s, the arc length along the curve, and solve for ϕ and z as funcition of s.

(c) Suppose the cylinder surface to be cut along a line parallel to its symmetry axis, and the cut surface then flattened out onto a table. Draw a diagonal line on that flattened surface and then re-assemble the cylinder. Determing if the line you drew (now a curve, of course) is an extremum path on the surface of the cylinder.

Solution

(a)

柱坐标 $ds = \sqrt{(d\rho)^2 + (\rho\, d\phi)^2 + (dz)^2} = \sqrt{\dot{\rho}^2 + \left(\rho\,\dot{\phi}\right)^2 + \dot{z}^2}\; d\beta$

现在 $\rho = a$, $ds = \sqrt{\left(a\,\dot{\phi}\right)^2 + \dot{z}^2}\; d\beta$

$l = \int ds = \int_A^B \sqrt{\left(a\,\dot{\phi}\right)^2 + \dot{z}^2}\; d\beta \implies f = \sqrt{\left(a\,\dot{\phi}\right)^2 + \dot{z}^2}$,

现在有两个变量，自然地，不涉及约化不约化问题，就是一个简单的两变量求变分问题

这样理解，就能很自然地利用欧拉-拉格朗日方程

$k = 1$ 时：$\dfrac{d}{d\beta}\dfrac{\partial f}{\partial \dot{\phi}} - \dfrac{\partial f}{\partial \phi} = 0$ （ϕ 相当于可遗坐标） $\implies \dfrac{a^2\,\dot{\phi}}{f} = c_1\, a$

$k = 2$ 时：$\dfrac{d}{d\beta}\dfrac{\partial f}{\partial \dot{z}} - \dfrac{\partial f}{\partial z} = 0$ （z 也相当于可遗坐标） $\implies \dfrac{\dot{z}}{f} = c_2$

(*b*)

现在，取 $\beta = s$，从出发点到当前点弧长。这有个好处，就是 $f = \dfrac{ds}{d\beta} = 1$

所以：$a\,\dot{\phi} = c_1$ 和 $\dot{z} = c_2$ 解得：$\phi = \dfrac{c_1}{a}\, s + \phi_0$, $z = c_2\, s + z_0$ 其中 ϕ_0, z_0 有初始坐标确定

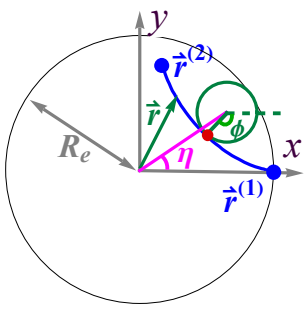
(*c*)

令 $x' = a\,\phi$ 则最短线的解可写为：$\begin{cases} x' = c_1\, s + x'_0 \\ z = c_2\, s + z_0 \end{cases}$ 这是平面中的直线方程

故：被切割平面中的直线就是柱面的短程线。

★ J05.09 （荣誉课程同学完成）

Suppose that a rail car moves without friction through a tunnel burrowed into the earth. It starts from the rest, and moves entirely under the influence of the nonuniform gavitivity inside the earth, assumed a sphere of radius R_e with uniform density. The gavitational constant is G and mass of the earth M. Ignore the rotation of the earth. Assume the tunnel lies in the x-o-y plane, the origin of coordinates is at center of the earth, and $\vec{e}_1$ points directly toward the point of entry.



(*a*) Using the general parametric methods（一般参数法）, write the Euler-Lagrange equations for the path that extremizes the transit time T from the entry point to the point $\vec{r}^{(2)} = \left(x^{(2)},\, y^{(2)},\, 0\right)$.

(*b*) By choosing the parameter β as η after the Euler-Lagrange equations are written, show the solution to these equations is, for suitable choice of R, a, given by

$$x = (R - a)\cos\eta + a\cos\left(\frac{R-a}{a}\,\eta\right), \qquad y = (R - a)\sin\eta - a\sin\left(\frac{R-a}{a}\,\eta\right)$$

which are the equations of a hypocycloid, the line traced out by a point on the circumference of a circle of radius a that is rolling without slipping along the inside of a circle of radius $R > a$.

The parameter η here is the plane-polar angle of the center of the rolling circle.

(*c*) Suppose that the far end of the extremum tunnel is back at the surface of the earth. If D is the distance along the surface of the earth between the entry and exit points, what is the greatest depth reached by the tunnel? How long did the trip take?

(*d*) Now rewrite the Euler-Lagrange equations with the choice $\beta = s$,

the arc length along the unvaried path. Write a single vector equation.

本题可借助 *Mathematica* 进行一些繁琐的计算。几行代码，可省去也许是几十分钟。

Solution

在地球内部任意位置 $\vec{r}$ 处的势能：$V(r) = \dfrac{m\,M\,G}{2\,R_e^3}\left(r^2 - R_e^2\right)$ （设地球表面势能为 0）

故此处车厢速度：$v = \sqrt{\dfrac{M\,G}{R_e^3}}\,\left(R_e^2 - r^2\right)^{1/2}$

时间极值等于线积分 I 极值：$I = \displaystyle\int_{\beta_1}^{\beta_2} \frac{d\,s}{\left(R_e^2 - r^2\right)^{1/2}} = \int_{\beta_1}^{\beta_2} \frac{\sqrt{\dot{x}^2 + \dot{y}^2}\;d\,\beta}{\left(R_e^2 - r^2\right)^{1/2}}$

被积函数：$f = \left(\dfrac{\dot{x}^2 + \dot{y}^2}{R_e^2 - r^2}\right)^{1/2}$

Euler-Lagrane 方程：$\dfrac{d}{d\,\beta}\left(\dfrac{\partial f}{\partial \dot{x}_k}\right) - \dfrac{\partial f}{\partial x_k} = 0, \quad x_k = x,\ y \ \text{ for } \ k = 1,\ 2$

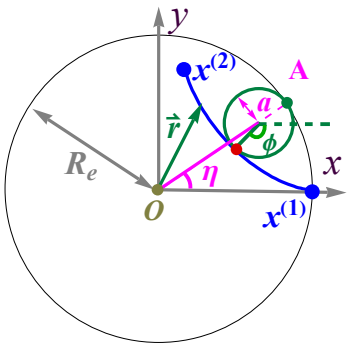
借助 *Mathematica* 可求得Euler-Lagrane 方程

$$\frac{d}{d\,\beta}\left(\frac{\dot{x}}{\sqrt{\left(\dot{x}^2 + \dot{y}^2\right)}\,\sqrt{\left(R_e^2 - r^2\right)}}\right) - \frac{x\,\sqrt{\left(\dot{x}^2 + \dot{y}^2\right)}}{\left(R_e^2 - r^2\right)^{3/2}} = 0,$$ 另一个方程类似

```
f[x_, y_, xd_, yd_] = (xd^2 + yd^2)/(R1^2 - (x^2 + y^2))^(1/2);
D[f[x, y, xd, yd], x]
D[f[x, y, xd, yd], xd]
```

$$\frac{x\left(xd^2 + yd^2\right)}{\left(R1^2 - x^2 - y^2\right)^2\sqrt{\dfrac{xd^2 + yd^2}{R1^2 - x^2 - y^2}}}$$

$$\frac{xd}{\left(R1^2 - x^2 - y^2\right)\sqrt{\dfrac{xd^2 + yd^2}{R1^2 - x^2 - y^2}}}$$



内摆线方程：见图

小圆圆心：$\begin{cases} x_0 = (R-a)\cos\eta \\ y_0 = (R-a)\sin\eta \end{cases}$

摆线上红点：$\begin{cases} x = x_0 + a\cos\phi \\ y = y_0 - a\sin\phi \end{cases}$ 图中 $\phi > 90°$

η 与 ϕ 的关系：$R\eta = a(\eta+\phi)$ （假设起始点在 x 轴，对应 $\phi = 0$）

摆线方程：$\begin{cases} x = x_0 + a\cos\phi = (R-a)\cos\eta + a\cos\left(\dfrac{R-a}{a}\eta\right) \\ y = y_0 - a\sin\phi = (R-a)\sin\eta - a\sin\left(\dfrac{R-a}{a}\eta\right) \end{cases}$

以下证明摆线方程满足 Euler-Lagrane 方程

$$\frac{d}{d\beta}\left(\frac{\dot{x}}{\sqrt{(\dot{x}^2+\dot{y}^2)}\,\sqrt{(R_e^2-r^2)}}\right) - \frac{x\sqrt{(\dot{x}^2+\dot{y}^2)}}{(R_e^2-r^2)^{3/2}} = 0$$

取 $\beta = \eta$，借助 Mathematica 可验证：$R = R_1$ 时，摆线满足 Euler-Lagrane 方程

```
x = (R - a) Cos[η] + a Cos[(R - a)/a η];
y = (R - a) Sin[η] - a Sin[(R - a)/a η];
xd = D[x, η];   yd = D[y, η];
r2 = x^2 + y^2;
rd2 = xd^2 + yd^2;
eq1 = D[xd/(Sqrt[xd^2 + yd^2] Sqrt[R1^2 - r2]), η] - x Sqrt[xd^2 + yd^2]/(R1^2 - r2)^(3/2);
eq2 = D[yd/(Sqrt[xd^2 + yd^2] Sqrt[R1^2 - r2]), η] - y Sqrt[xd^2 + yd^2]/(R1^2 - r2)^(3/2);
```

```
Simplify[eq1]
Simplify[eq2];
```

$$\left(\left(2a^2-3aR+R^2\right)\left(R^2-R1^2\right)\mathrm{Sin}\left[\frac{R\eta}{2a}\right]\mathrm{Sin}\left[\eta-\frac{R\eta}{2a}\right]\right)\Big/\left(\sqrt{2}\,a\sqrt{-(a-R)^2\left(-1+\mathrm{Cos}\left[\frac{R\eta}{a}\right]\right)}\left(-2a^2+2aR-R^2+R1^2+2a(a-R)\mathrm{Cos}\left[\frac{R\eta}{a}\right]\right)^{3/2}\right)$$

$$-4a(a-R)\,\mathrm{Sin}\left[\frac{R\eta}{2a}\right]^2$$

接着，求出口的 η

```
Factor[Simplify[R^2 - r2]]
Factor[Simplify[rd2]]
```

$$-4a(a-R)\,\mathrm{Sin}\left[\frac{R\eta}{2a}\right]^2$$

$$4(a-R)^2\,\mathrm{Sin}\left[\frac{R\eta}{2a}\right]^2$$

作为出口，$R^2-(x^2+y^2)=0 \implies -4a(a-R)\sin\left(\frac{R\eta}{2a}\right)^2=0 \implies \frac{\eta R}{2a}=\pi \implies \eta=\frac{2a\pi}{R}$

另一方面，出口距离进口为 D，故：$R\eta=D$　　联立求得：$a=\frac{D}{2\pi}$

最深点为 $2a$，即 $d_{\max}=\frac{D}{\pi}$

所需时间：$T=\sqrt{\frac{R_e^3}{MG}}\,I$，　由上代码求得：$\sqrt{\dot{x}^2+\dot{y}^2}=2(R-a)\sin\left(\frac{R\eta}{2a}\right)$

$$\text{先求}\quad I=\int_{\eta_1}^{\eta_2}\frac{\sqrt{\dot{x}^2+\dot{y}^2}\,d\eta}{\left(R_e^2-r^2\right)^{1/2}}=\int_{\eta_1}^{\eta_2}\frac{2(R-a)\sin\left(\frac{R\eta}{2a}\right)d\eta}{2\sqrt{a(R-a)}\,\sin\left(\frac{R\eta}{2a}\right)}=\sqrt{\frac{R-a}{a}}\,\frac{2a\pi}{R}=\frac{D}{R}\sqrt{\frac{2\pi R}{D}-1}$$

$$T=\sqrt{\frac{R^3}{MG}}\,I\xrightarrow[\text{重力加速度 }g]{MG/R^2=g}\sqrt{\frac{R}{g}}\,I=\sqrt{2\pi}\sqrt{\frac{2\pi R}{g}}\sqrt{\frac{D}{2\pi R}\left(1-\frac{D}{2\pi R}\right)}$$

$D=5000\,\mathrm{km}$ 时，由 $2\pi R\approx 40\,000\,\mathrm{km}$　求得 $T\approx 28\,\mathrm{min}$

若取 $\beta=s$ 弧长，则 $\sqrt{(\dot{x}^2+\dot{y}^2)}=\frac{ds}{ds}=1$

Euler-Lagrange 方程

$$\frac{d}{d\beta}\left(\frac{\dot{x}}{\sqrt{(\dot{x}^2+\dot{y}^2)}\ \sqrt{(R_e^2-r^2)}}\right)-\frac{x\sqrt{(\dot{x}^2+\dot{y}^2)}}{(R_e^2-r^2)^{3/2}}=0$$

可写成矢量形式：

$$\frac{d}{d\beta}\left(\frac{\vec{\tau}}{\sqrt{R^2-r^2}}\right)-\frac{\vec{r}}{(R_e^2-r^2)^{3/2}}=0，其中\vec{\tau}为曲线切向：\vec{\tau}=\frac{dx}{ds}\vec{e}_x+\frac{dy}{ds}\vec{e}_y$$

当然，如果熟悉空间曲线的 Serret-Frenet 理论，从这个方程也可导出沿内摆线时间取极值。

Mathematica 中可以调用 FrenetSerretSystem 帮助求解。

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?FrenetSerretSystem
```

FrenetSerretSystem[{x_1, …, x_n}, t] gives the generalized curvatures and Frenet–Serret basis for the parametric curve $x_i[t]$.
FrenetSerretSystem[{x_1, …, x_n}, t, *chart*] interprets the x_i as coordinates in the specified coordinate chart.

★ J15.02 （诺特定理）涉及周三(11.01)的内容

Suppose that a system of N point masses has a potential function

$$U(\vec{r}_1,\vec{r}_2,\ldots,\vec{r}_N)=\sum_{n=1}^{N}\sum_{n'=1}^{n-1}f(\vec{r}_n-\vec{r}_{n'})$$

Use Noether's theorem to prove that the total momentum vector $\vec{P}$ is a constant of the motion. Is total angular momentum $\vec{J}$ also a constant of the motion?

Solution

扩展的拉格朗日量：$\mathcal{L}=\frac{1}{2\dot{t}}\sum_n m_n\dot{\vec{r}}_n\cdot\dot{\vec{r}}_n-\dot{t}\sum_{n=1}^{N}\sum_{n'=1}^{n-1}f(\vec{r}_n-\vec{r}_{n'})$

平动

$\vec{r}_n\longrightarrow\vec{r}_n+\vec{m}\,\delta a$　其中 $\vec{m}$ 为常矢量，与 n，β 都无关（相当于每个质点位置都平移相同的一个小量）

$\delta\vec{r}_n=\vec{m}\,\delta a,\quad \delta\dot{\vec{r}}_n=0$

$\vec{r}_n-\vec{r}_{n'}\longrightarrow\vec{r}_n+\vec{m}\,\delta a-(\vec{r}_{n'}+\vec{m}\,\delta a)=\vec{r}_n-\vec{r}_{n'}$

所以 拉格朗日量的第一、第二项都不因 变分 $\vec{r}_n\longrightarrow\vec{r}_n+\vec{m}\,\delta a$ 而改变，即 $\delta\mathcal{L}=0$

存在守恒量：$\sum_{k=1}^{3N}p_k\,\delta q_k=\sum_n\vec{p}_n\cdot\vec{m}\,\delta a=\vec{P}\cdot\vec{m}\,\delta a$

因为 $\vec{m}$ 是任意的，故 总动量 $\vec{P}$ 是守恒量

转动

为简单起见，绕 z 轴转

可构造变换：$\begin{cases}x_i'=x_i\cos\theta-y_i\sin\theta\\ y_i'=x_i\sin\theta+y_i\cos\theta\\ z_i'=z_i\end{cases}$　是体系转动，坐标系不动

看无穷小变换：θ 很小，$\cos\theta=1$，$\sin\theta=\theta$

变换为：$\begin{cases}x_i'=x_i-y_i\theta\\ y_i'=x_i\theta+y_i\\ z_i'=z_i\end{cases}$　$\begin{cases}\delta x_i=-y_i\theta\\ \delta y_i=x_i\theta\\ \delta z_i=0\end{cases}$　$\begin{cases}\delta\dot{x}_i=-\dot{y}_i\theta\\ \delta\dot{y}_i=\dot{x}_i\theta\\ \delta\dot{z}_i=0\end{cases}$

θ 相当于δa，是小量

看 $\delta\mathcal{L}$

第一项：$\delta\left(\dot{\vec{r}}_n\cdot\dot{\vec{r}}_n\right)=2\,\dot{\vec{r}}_i\cdot\delta\,\dot{\vec{r}}_n=2\,\dot{\vec{r}}_i\cdot\left(-\dot{y}_i\,\theta\,\vec{e}_x+\dot{x}_i\,\theta\,\vec{e}_y\right)$

$$=2\left(\dot{x}_i\,\vec{e}_x+\dot{y}_i\,\vec{e}_y+\dot{z}_i\,\vec{e}_z\right)\cdot\left(-\dot{y}_i\,\theta\,\vec{e}_x+\dot{x}_i\,\theta\,\vec{e}_y\right)=0$$

第一项满足不变性

第二项：$f\left(\vec{r}_n-\vec{r}_{n'}\right)$ 与相对取向有关

为说明角动量未必守恒，假设势能 $f\left(\vec{r}_n-\vec{r}_{n'}\right)=\vec{e}_x\cdot\left(\vec{r}_n-\vec{r}_{n'}\right)$

$$\delta\,f\left(\vec{r}_i-\vec{r}_j\right)=\frac{\partial\,f\left(\vec{r}_i-\vec{r}_j\right)}{\partial\,\vec{r}_i}\cdot\delta\,\vec{r}_i$$

$$=\frac{\partial\left[\vec{e}_x\cdot\left(\vec{r}_i-\vec{r}_j\right)\right]}{\partial\,\vec{r}_i}\cdot\left(-y_i\,\theta\,\vec{e}_x+x_i\,\theta\,\vec{e}_y\right)$$

$$=\vec{e}_x\cdot\left(-y_i\,\theta\,\vec{e}_x+x_i\,\theta\,\vec{e}_y\right)=-y_i\,\theta\neq 0$$

故 $\delta\mathcal{L}\neq 0$，且不等于 $\dfrac{d\mathcal{F}(q)}{d\,\beta}\,\delta\,a$ 形式，无法构造一个守恒量。总角动量**未必**守恒。